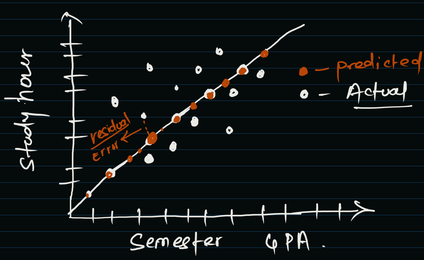
Linear Regression

Regression is supervised learning task used to Predict a continious numerical output based on one or more feature.



Model Equation :

[$\hat{y} = w_1 x + b$] — Prediction Model

[$Y = w_1 x + b + \epsilon$] — Actual Model

Y = The actual ground truth value

$\hat{Y}$ = Model is estimated Predicted

w_1 = weight (importance/scaling factor) for Feature x_1

b = The bias, it remains the baseline prediction when $x_1 = 0$

Error : The difference between our prediction and the actual truth for a single data point.

$$e = \hat{y} - y$$

Why Squared Error:

(i) It insure negative and positive error do not cancel each other out

(ii) It heavily penalizes large errors, forcing the model to correct them.

Mean Squared Error (MSE) / cost function (J)

$$J = \frac{1}{n} \sum_{i=1}^{n} (\hat{y}^{(i)} - y^{(i)})^2$$

[Gradient Descent]

- Learning rate (α) : Determines the size of the steps you take
  - To small :- Learning is extremely slow
  - To large : You might overshoot the valley and fail to converge.
- Update Rule :

Parameter ← parameter − α * gradient.

We substract the gradient because the gradient points in the direction of the steepest ascent. To minimize loss, we must move in the opposite direction (downhill).

[Partial Derivates] :

We need to know how changing w_1 and b affect the overall loss J. we do this by taking the partial derivative of J with respect of each parameter.

Starting Equation :

$$J = \frac{1}{2n} \sum_{i=1}^{n} (w_1 x^{(i)} + b - y^{(i)})^2$$

Chain rule :

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x).$$

(i) Derivatives with respect to (w_1) :

$$\frac{dJ}{dw_1} = \frac{1}{2n} \sum (w_1 x + b - y) \cdot \frac{d}{dw_1} (w_1 x + b - y)$$

- The derivative of inner function ($w_1 x + b - y$) with respect to w_1 is Just x (since b and y are treated as constant and the derivative of $w_1 x$ is x)

$$\frac{dJ}{dw_1} = \frac{1}{2n} \sum_{i=1}^{n} (\hat{y}^{(i)} - y^{(i)}) * x^{(i)}$$

(ii) Derivative with respect to (b) :

$$\frac{dJ}{db} = \frac{1}{2n} \sum_{i=1}^{n} (w_1 x + b - y) \cdot \frac{d}{db} (w_1 x + b - y)$$

- The derivative of inner function ($w_1 x + b - y$) with respect to b is exactly 1 (since $w_1 x$ and y are constant)

$$\frac{dJ}{db} = \frac{1}{2n} \sum_{i=1}^{n} (\hat{y}^{(i)} - y^{(i)})$$

Note:- you may notice weight derivates contains x, while the basis derivates does not. ?

Because of their placement in the model equation

$\hat{y} = w_1 x + b$. The weight w_1 is multiplied by the input feature x, meaning its impact on the prediction scales with x. The bias b is a standalone constant, so its rate of change is independent of the input feature.

Gradient Descent update :

$$w_1 \leftarrow w_1 - \alpha \left(\frac{1}{2n} \sum_{i=1}^{n} (\hat{y}^{(i)} - y^{(i)}) x^{(i)} \right)$$

$$b \leftarrow b - \alpha \left(\frac{1}{n} \sum_{i=1}^{n} (\hat{y}^{(i)} - y^{(i)}) \right)$$

[Summary :]

Prediction → Error → loss → partial Derivatives → Gradient → parameter update → Repeat.